Psych 250a — Week 1 Assignment. Describing Data

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Question 1: Import and Summarize the Data

```r
library(ggplot2)
library(psych)
library(summarytools)

filepath <-
 ↪  "https://raw.githubusercontent.com/craigenders/psych250a/main/data/assignment1.csv"
assign1 <- read.csv(filepath, stringsAsFactors = TRUE)

dfSummary(assign1)
```

```
Data Frame Summary
assign1
Dimensions: 297 x 11
Duplicates: 0

----------------------------------------------------------------------------------------------------
 ↪  ----------------
No   Variable    Stats / Values           Freqs (% of Valid)    Graph                Valid
 ↪  Missing
---- ----------- ------------------------ --------------------- --------------------
 ↪  ---------- ---------
1    ID          Mean (sd) : 149 (85.9)   297 distinct values   : : : : : .          297
 ↪  0
     [integer]   min < med < max:         (Integer sequence)    : : : : : :
      ↪  (100.0%)   (0.0%)
                 1 < 149 < 297                                  : : : : : :
                 IQR (CV) : 148 (0.6)                           : : : : : :
                                                                : : : : : :

2    Female      Min  : 0                 0 : 189 (63.6%)       IIIIIIIIIII          297
 ↪  0
     [integer]   Mean : 0.4               1 : 108 (36.4%)       IIIIIII
      ↪  (100.0%)   (0.0%)
                 Max  : 1

3    ELS         Min  : 0                 0 : 180 (60.6%)       IIIIIIIIIII          297
 ↪  0
     [integer]   Mean : 0.4               1 : 117 (39.4%)       IIIIIII
      ↪  (100.0%)   (0.0%)
                 Max  : 1

4    Age         Mean (sd) : 45.6 (5.7)   32 distinct values           :           297
 ↪  0
     [integer]   min < med < max:                                    : :
      ↪  (100.0%)   (0.0%)
                 26 < 46 < 62                                       : :
                 IQR (CV) : 7 (0.1)                               : : : :
                                                                . : : : : .

5    CRP         Mean (sd) : 0.2 (1.3)    280 distinct values   :                    297
 ↪  0
     [numeric]   min < med < max:                               :
      ↪  (100.0%)   (0.0%)
```

```
                -1.2 < -0.1 < 9.7                          : :
                IQR (CV) : 1 (5.7)                         : :
                                                           : : : .

6    IL6        Mean (sd) : -0.2 (0.8)   273 distinct values     : .              297
 ↪  0
     [numeric]  min < med < max:                                 : :
      ↪  (100.0%)   (0.0%)
                -1.7 < -0.3 < 4.8                                : :
                IQR (CV) : 0.8 (-3.5)                            : :
                                                           . : : : .

7    TNF        Mean (sd) : 0.5 (0.9)    278 distinct values          . : .       297
 ↪  0
     [numeric]  min < med < max:                                        : : :
      ↪  (100.0%)   (0.0%)
                -2.8 < 0.5 < 3.3                                  . : : :
                IQR (CV) : 1.3 (1.8)                              : : : : :
                                                                . : : : : : : :

8    IFN        Mean (sd) : 0.3 (0.8)    277 distinct values     :                297
 ↪  0
     [numeric]  min < med < max:                                 :
      ↪  (100.0%)   (0.0%)
                -1 < 0.1 < 3.2                                   : : .
                IQR (CV) : 0.9 (2.5)                             : : :
                                                           : : : : : . . .

9    DPDD       Mean (sd) : 6.3 (4.5)    280 distinct values   :                297
 ↪  0
     [numeric]  min < med < max:                                 : :
      ↪  (100.0%)   (0.0%)
                0 < 5.5 < 29.1                                   : :
                IQR (CV) : 4.7 (0.7)                             : :
                                                                 : : : .

10   HDD        Mean (sd) : 2.1 (1.9)    0 : 65 (21.9%)        IIII               297
 ↪  0
     [integer]  min < med < max:         1 : 77 (25.9%)        IIIII
      ↪  (100.0%)   (0.0%)
                0 < 2 < 7                2 : 49 (16.5%)        III
                IQR (CV) : 2 (0.9)       3 : 47 (15.8%)        III
                                         4 : 28 ( 9.4%)        I
                                         5 :  6 ( 2.0%)
                                         6 :  8 ( 2.7%)
                                         7 : 17 ( 5.7%)        I

11   HvyDrink   Min  : 0                0 :  65 (21.9%)       IIII               297
 ↪  0
     [integer]  Mean : 0.8               1 : 232 (78.1%)       IIIIIIIIIIIIII
      ↪  (100.0%)   (0.0%)
                Max  : 1
-------------------------------------------------------------------------------
 ↪  ---------------
```

Question 2: Convert ELS and Female to Factors

Your code:

```
assign1$ELS <- factor(assign1$ELS,
                          levels = c(0, 1),
                          labels = c("No early life stress", "Early life stress"))

assign1$Female <- factor(assign1$Female,
                          levels = c(0, 1),
                          labels = c("Male", "Female"))

freq(assign1)
```

```
Frequencies
assign1$Female
Type: Factor

               Freq   % Valid   % Valid Cum.   % Total   % Total Cum.
------------ ------ --------- -------------- --------- --------------
       Male    189     63.64          63.64     63.64          63.64
     Female    108     36.36         100.00     36.36         100.00
      <NA>      0                                0.00         100.00
      Total    297    100.00         100.00    100.00         100.00

assign1$ELS
Type: Factor

                           Freq   % Valid   % Valid Cum.   % Total   % Total Cum.
-------------------------- ------ --------- -------------- --------- --------------
     No early life stress    180     60.61          60.61     60.61          60.61
        Early life stress    117     39.39         100.00     39.39         100.00
                    <NA>      0                                0.00         100.00
                   Total    297    100.00         100.00    100.00         100.00

assign1$HDD
Type: Integer

              Freq   % Valid   % Valid Cum.   % Total   % Total Cum.
---------- ------ --------- -------------- --------- --------------
         0     65     21.89          21.89     21.89          21.89
         1     77     25.93          47.81     25.93          47.81
         2     49     16.50          64.31     16.50          64.31
         3     47     15.82          80.13     15.82          80.13
         4     28      9.43          89.56      9.43          89.56
         5      6      2.02          91.58      2.02          91.58
         6      8      2.69          94.28      2.69          94.28
         7     17      5.72         100.00      5.72         100.00
      <NA>      0                                0.00         100.00
     Total    297    100.00         100.00    100.00         100.00

assign1$HvyDrink
Type: Integer
```

	Freq	% Valid	% Valid Cum.	% Total	% Total Cum.
0	65	21.89	21.89	21.89	21.89
1	232	78.11	100.00	78.11	100.00
<NA>	0			0.00	100.00
Total	297	100.00	100.00	100.00	100.00

Question 3: Inflammation Sum Score

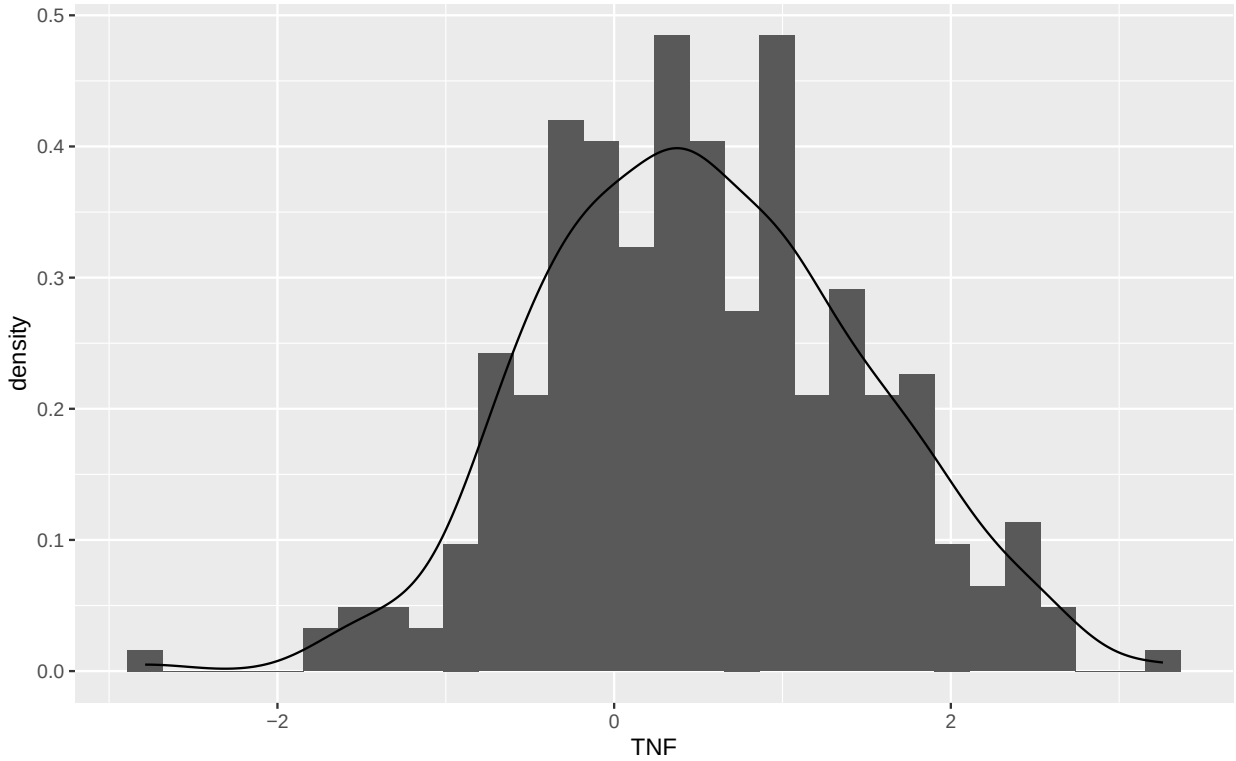
Your code:

```
assign1$Inflammation <- rowSums(
assign1[, c("CRP",
"IL6",
"TNF",
"IFN")]
)
```

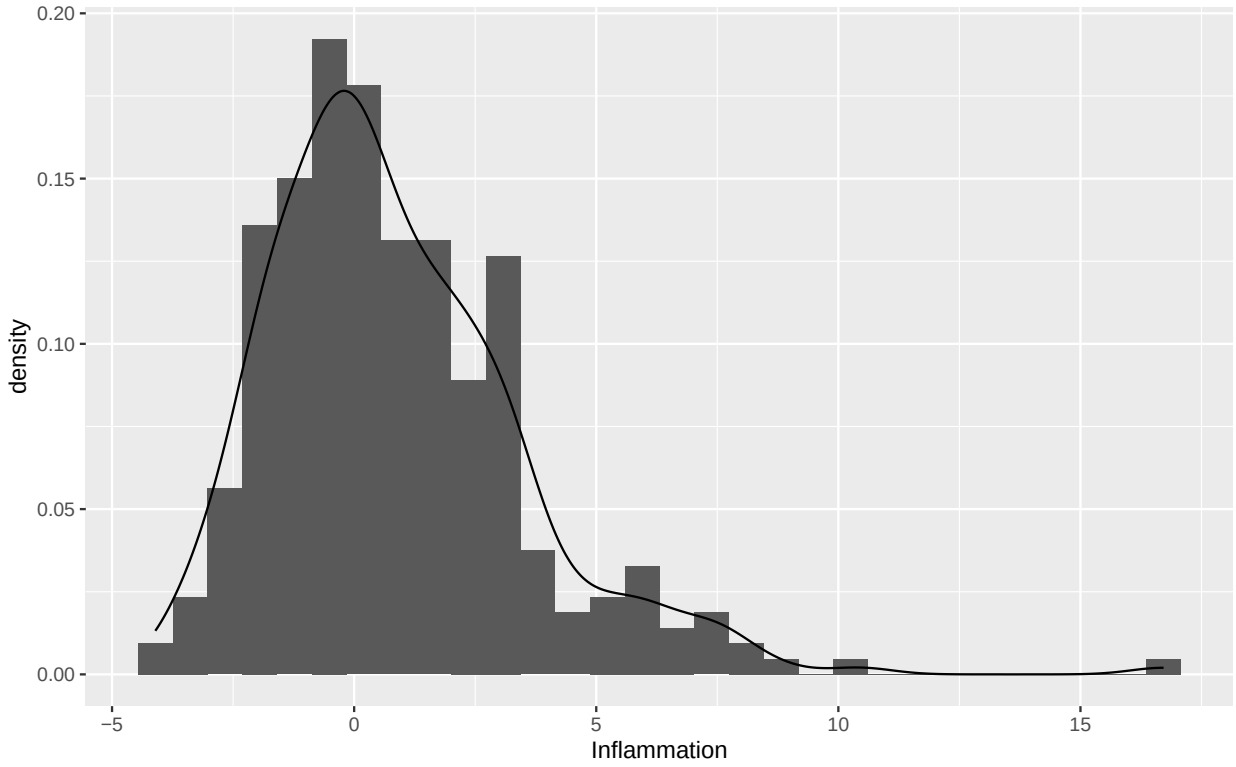
Question 4: Histograms with Density Curves

Your code:

```
# TNF
ggplot(assign1, aes(x = TNF)) +
  geom_histogram(aes(y = after_stat(density))) +
  geom_density()
```



```r
# Inflammation
ggplot(assign1, aes(x = Inflammation)) +
  geom_histogram(aes(y = after_stat(density))) +
  geom_density()
```



Question 5: TNF Distribution Shape

Based on the histogram and density curve, the distribution of TNF appears approximately symmetric and bell-shaped, with most observations concentrated around the center and fewer observations toward either end. The left and right sides of the distribution are relatively balanced, and neither tail appears to be substantially longer than the other. I assume that the skewness and kurtosis values would be near zero.

Question 6: Inflammation Distribution Shape

Based on the histogram and density curve, the distribution of Inflammation appears positively skewed, with most observations concentrated at relatively low values and a long tail extending toward higher values. A small number of extreme observations contribute to the positively skewed shape. Moreover, the distribution also shows relatively heavy right tail, which may indicate to relatively high kurtosis.

Question 7: Descriptive Statistics

Your code:

```
describe(assign1)
```

```
             vars   n   mean    sd median trimmed    mad   min    max  range
ID              1 297 149.00 85.88 149.00  149.00 109.71  1.00 297.00 296.00
Female*         2 297   1.36  0.48   1.00    1.33   0.00  1.00   2.00   1.00
ELS*            3 297   1.39  0.49   1.00    1.37   0.00  1.00   2.00   1.00
Age             4 297  45.59  5.66  46.00   45.74   5.93 26.00  62.00  36.00
CRP             5 297   0.23  1.30  -0.12    0.02   0.67 -1.19   9.73  10.92
IL6             6 297  -0.22  0.76  -0.33   -0.31   0.55 -1.68   4.81   6.48
TNF             7 297   0.53  0.94   0.46    0.51   0.94 -2.79   3.26   6.05
IFN             8 297   0.31  0.77   0.12    0.21   0.66 -0.99   3.21   4.20
DPDD            9 297   6.29  4.53   5.45    5.74   3.48  0.00  29.14  29.14
HDD            10 297   2.10  1.93   2.00    1.83   1.48  0.00   7.00   7.00
HvyDrink       11 297   0.78  0.41   1.00    0.85   0.00  0.00   1.00   1.00
Inflammation   12 297   0.85  2.68   0.37    0.59   2.40 -4.11  16.72  20.82
              skew kurtosis   se
ID            0.00    -1.21 4.98
Female*       0.56    -1.69 0.03
ELS*          0.43    -1.82 0.03
Age          -0.27     0.49 0.33
CRP           3.33    16.40 0.08
IL6           2.43    10.42 0.04
TNF           0.06    -0.05 0.05
IFN           1.21     1.53 0.04
DPDD          1.45     3.16 0.26
HDD           1.00     0.37 0.11
HvyDrink     -1.35    -0.17 0.02
Inflammation  1.36     4.06 0.16
```

Skewness and excess kurtosis for `TNF` and `Inflammation`:

TNF) skewness: **0.06** / excess kurtosis: **-0.05**
Inflammation) skewness: **1.36** / excess kurtosis: **4.06**

Question 8: TNF Skewness and Kurtosis

The skewness of TNF is 0.06, which is close to zero and indicates that the distribution has little asymmetry. Its excess kurtosis is -0.05, which is also close to zero and indicates that its kurtosis is similar to that of a normal distribution. These statistics support my interpretation that TNF is approximately symmetric and bell-shaped, consistent with histogram.

Question 9: Inflammation Skewness and Kurtosis

The skewness of inflammation is 1.36, indicating substantial positive skewness. This means that the distribution is asymmetric, with a longer right tail. Its excess kurtosis is 4.06, which is substantially greater than zero and suggests heavier tails and a greater tendency toward extreme observations, compared to a normal distribution. These also support my interpretation that inflammation departs from a normal distribution, particularly in its asymmetry and heavy right tail.

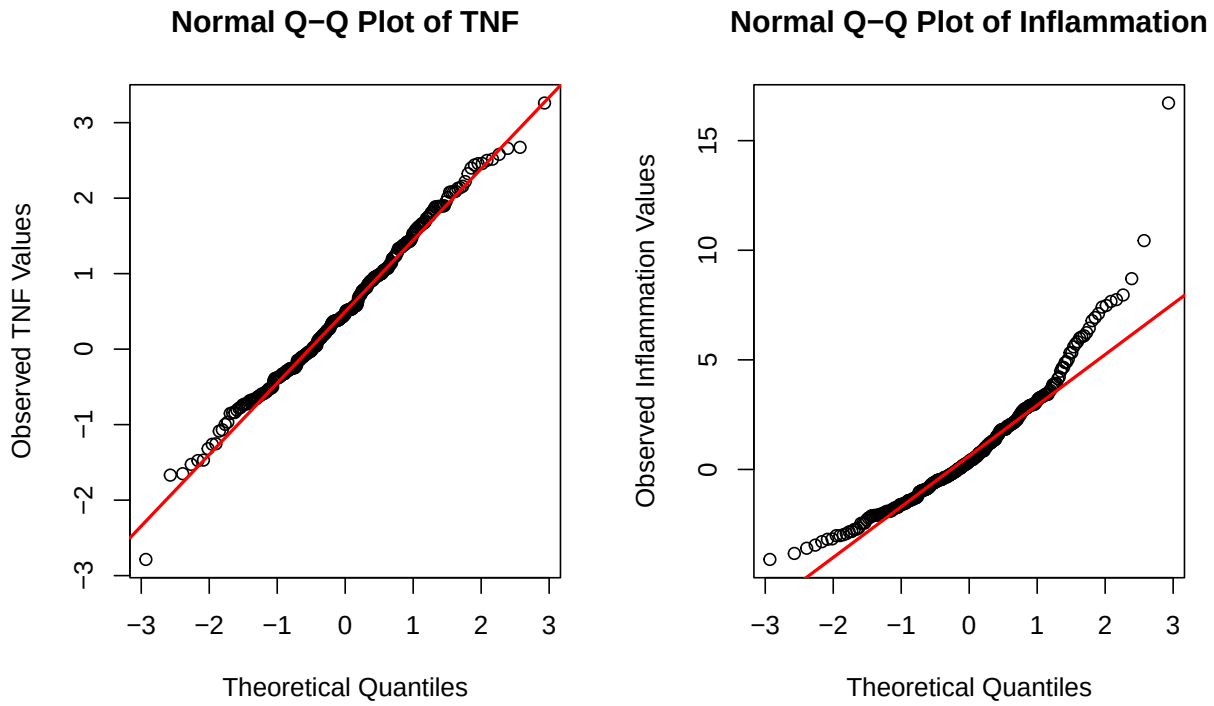
Question 10: AI Prompt for Q–Q Plots

AI's code:

```r
par(mfrow = c(1, 2))

qqnorm(assign1$TNF,
       main = "Normal Q-Q Plot of TNF",
       ylab = "Observed TNF Values")
qqline(assign1$TNF, col = "red", lwd = 2)

qqnorm(assign1$Inflammation,
       main = "Normal Q-Q Plot of Inflammation",
       ylab = "Observed Inflammation Values")
qqline(assign1$Inflammation, col = "red", lwd = 2)
```




```r
par(mfrow = c(1, 1))
```

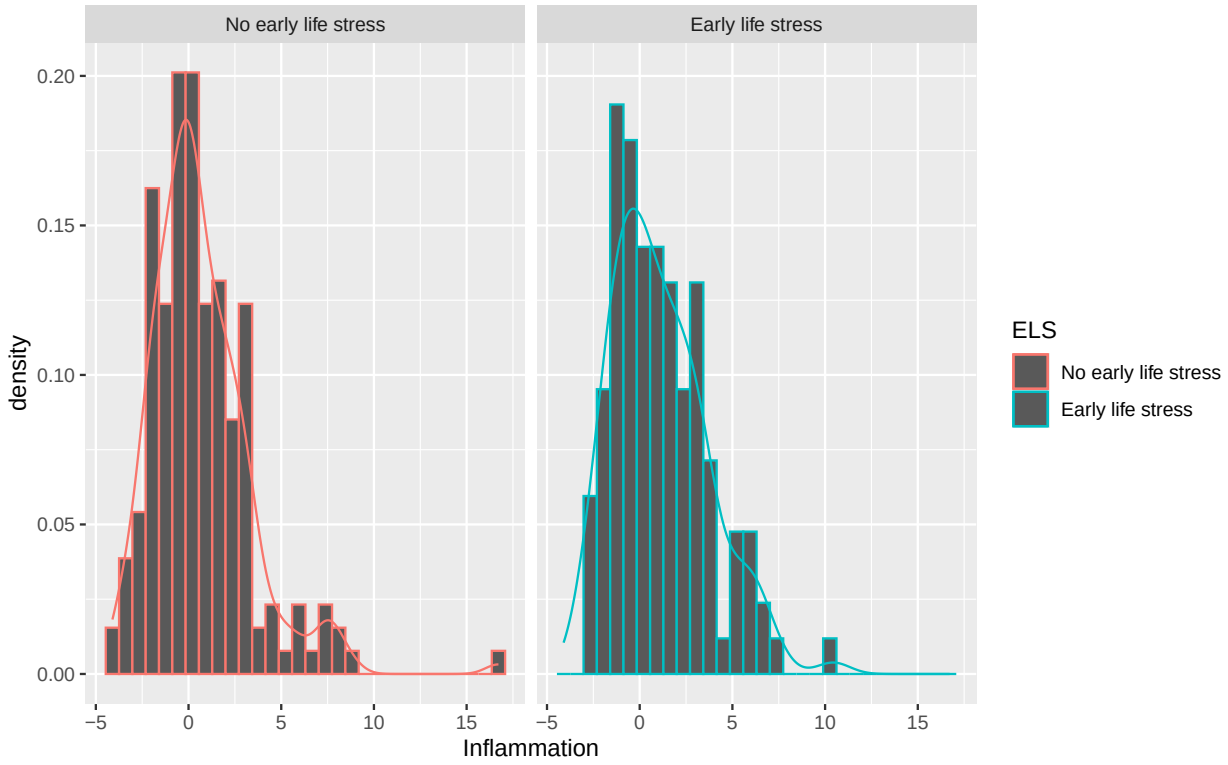
Question 11: Interpret the Q–Q Plot

In terms of the inflammation composite, the Q-Q plot shows that the data points at the upper end show substantial departures above the reference line. In other words, the largest observations are farther from the center than expected under the normality. At the lower end, the smallest observations also appear above the reference line, suggesting that they are closer to the center than expected under normality. Meanwhile, these results suggest high kurtosis, which is associated with heavy tails. In addition, the departures at the upper end are much more extreme than those at the lower end, indicating positive skewness. These features can also be observed in the histogram, which shows positive skewness and a heavy right tail.

Question 12: Side-by-Side Histograms by ELS

Your code:

```
ggplot(assign1, aes(x = Inflammation, colour = ELS)) +
  geom_histogram(aes(y = after_stat(density))) +
  geom_density() +
  facet_wrap(~ ELS)
```



Question 13: AI Suggestions to Improve the Graph

Which suggestion you selected and why it should improve the display:

I chose Option 2: Add group means or medians. This option retains the histogram and adds vertical lines using `geom_vline()` to indicate the mean value, allowing for better comparisons between groups. However, since it is hard to determine the exact mean value from the graph, I additionally asked the AI to add labels showing the mean and median values for each group. This gives me a better understanding of which group has higher values of central tendency.

The modified code:

```
library(dplyr)
library(ggplot2)

stats <- assign1 %>%
  group_by(ELS) %>%
  summarise(
```

```
    mean_inf = mean(Inflammation, na.rm = TRUE),
    median_inf = median(Inflammation, na.rm = TRUE)
  )

ggplot(assign1, aes(x = Inflammation)) +
  geom_histogram(
    aes(y = after_stat(density)),
    bins = 15,
    fill = "grey80",
    colour = "black"
  ) +

  # Mean
  geom_vline(
    data = stats,
    aes(xintercept = mean_inf),
    colour = "blue",
    linewidth = 1.2
  ) +

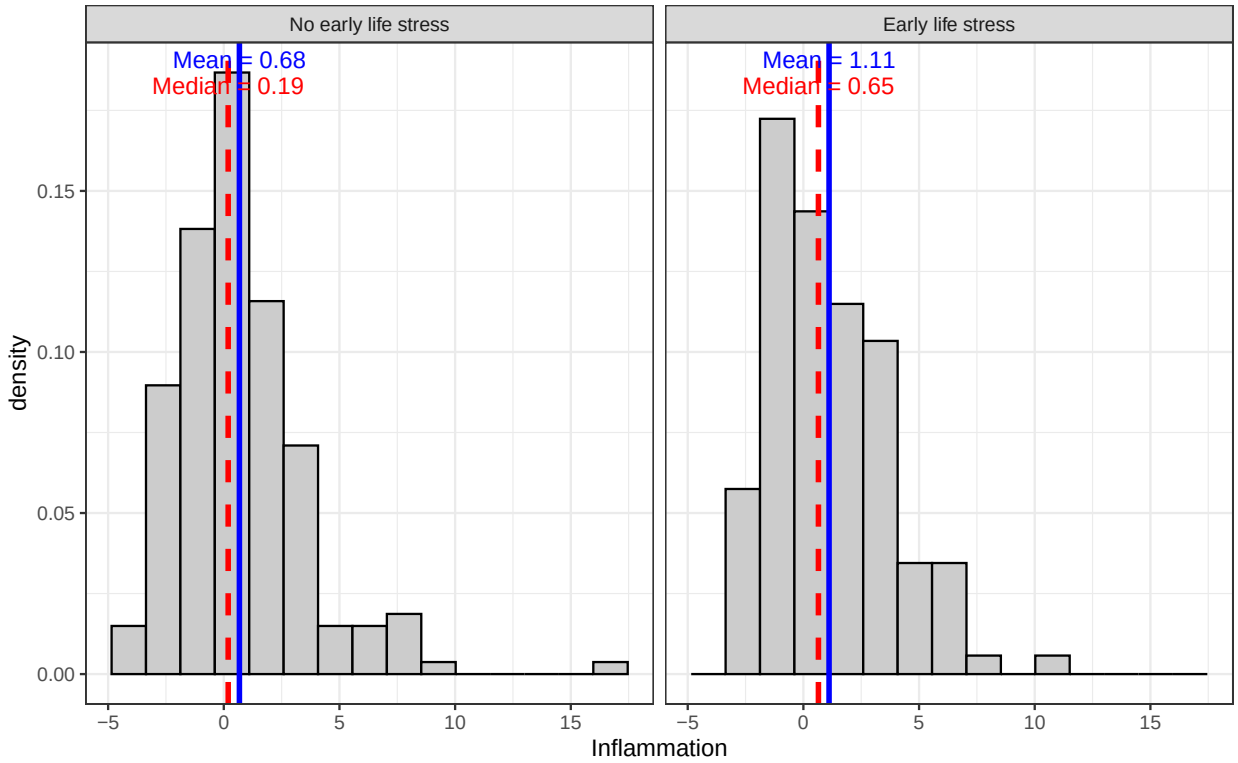
  # Median
  geom_vline(
    data = stats,
    aes(xintercept = median_inf),
    colour = "red",
    linewidth = 1.2,
    linetype = "dashed"
  ) +

  # Mean label
  geom_text(
    data = stats,
    aes(
      x = mean_inf,
      y = Inf,
      label = paste0("Mean = ", round(mean_inf, 2))
    ),
    colour = "blue",
    vjust = 1.5
  ) +

  # Median label
  geom_text(
    data = stats,
    aes(
      x = median_inf,
      y = Inf,
      label = paste0("Median = ", round(median_inf, 2))
    ),
    colour = "red",
    vjust = 3
  ) +

  facet_wrap(~ELS) +
```

```
  theme_bw()
```



Question 14: Conclusion About ELS and Inflammation

According to the mean and median values, early life stress group shows higher mean and median levels of inflammation, which may suggest a relationship between early life stress and higher levels of inflammation later in life. However, I cannot confidentially conclude that there is a significant difference between the two groups, especially in terms of their distributions. Both distributions appear positively skewed and overlap substantially.

Question 15: Chat Log

```
You said:
I'd like to draw normal Q-Q plots to examine the shape of distributions. I am currently
 ↪  using a data named with "assign1" and I'd like to work with the variables "TNF" and
 ↪  "Inflammation". I wanted to ask you to:
(1) provide R code that creates separate normal Q-Q plots for these two variables: "TNF"
 ↪  and "Inflammation", each with a reference line, and (2) explain what the axes and
 ↪  reference lines represent and how departures from the lines can provide evidence
 ↪  about distribution shape.

Copilot said:
To create separate normal Q-Q plots for TNF and Inflammation from your dataset assign1,
 ↪  you can use the base R functions qqnorm() and qqline():
```

```
# Q-Q plot for TNF
qqnorm(assign1$TNF,
       main = "Normal Q-Q Plot of TNF",
       ylab = "Observed TNF Values")
qqline(assign1$TNF, col = "red", lwd = 2)

# Q-Q plot for Inflammation
qqnorm(assign1$Inflammation,
       main = "Normal Q-Q Plot of Inflammation",
       ylab = "Observed Inflammation Values")
qqline(assign1$Inflammation, col = "red", lwd = 2)
If you'd like both plots displayed side-by-side:



par(mfrow = c(1, 2))

qqnorm(assign1$TNF,
       main = "Normal Q-Q Plot of TNF",
       ylab = "Observed TNF Values")
qqline(assign1$TNF, col = "red", lwd = 2)

qqnorm(assign1$Inflammation,
       main = "Normal Q-Q Plot of Inflammation",
       ylab = "Observed Inflammation Values")
qqline(assign1$Inflammation, col = "red", lwd = 2)

par(mfrow = c(1, 1))
Interpreting the Q-Q Plots
Axes
Horizontal (x) axis: Theoretical quantiles from a normal distribution.

These are the values we would expect if the data were perfectly normally distributed.
Vertical (y) axis: Sample quantiles (ordered observed values) from your variable (TNF or
 ↪  Inflammation).

Each point compares an observed value to the value expected under normality.

Reference Line
The line added by qqline() is a reference line representing the pattern expected if the
 ↪  data followed a normal distribution. The line is drawn using the first and third
 ↪  quartiles of the sample and corresponding theoretical normal quantiles.

What Departures from the Line Mean
Points close to the line throughout the plot

Evidence that the distribution is approximately normal.
Right tail bends upward above the line

Suggests a heavier upper tail than a normal distribution.
Often indicates positive (right) skewness.
Left tail bends downward below the line
```

```
Also consistent with positive skewness.
Left tail above the line and right tail below the line

Suggests negative (left) skewness.
Both tails far from the line in the same direction, creating an S-shape

Indicates tails that are heavier or lighter than normal.
Heavy tails (more extreme values) often appear with points below the line on the left and
 ↪ above the line on the right.
Light tails show the opposite pattern.
Extreme points far from the line

May indicate outliers.
Example Reporting Statement
The normal Q-Q plot for TNF was examined to assess normality. If the plotted points fall
 ↪ approximately along the reference line, the distribution can be considered
 ↪ approximately normal. Systematic departures from the line, particularly in the tails,
 ↪ would suggest skewness, heavy tails, or other deviations from normality. The same
 ↪ interpretation applies to the Inflammation variable.

This interpretation is often preferred over formal normality tests because it provides
 ↪ information about how the distribution departs from normality, not just whether it
 ↪ does.
```

```
You said:
Rplot01.png
Ok. let's move onto next task.
I'd like to figure out whether participants who report adverse early life stressors ( =
 ↪ 1) have different levels of inflammation than participants who report no adverse
 ↪ stressors ( = 0). In other words, I'd like to determine whether adverse early-life
 ↪ stressors are associated with inflammation later in life. Based on the data frame
 ↪ 'assign1' and variable 'Inflammation' and grouping variable 'ELS', I draw the
 ↪ histogram and density plot using ggplot2:
ggplot(assign1, aes(x = Inflammation, colour = ELS)) + geom_histogram(aes(y =
 ↪ after_stat(density))) + geom_density() + facet_wrap(~ ELS)
and I attached the result figure.
I'd like to ask you for suggesting ways to improve the graph so that differences between
 ↪ the two groups are easier to evaluate, while retaining histograms as the style of
 ↪ display.

Copilot said:
Your current figure is a reasonable start because the faceting separates the two ELS
 ↪ groups, but several design choices make comparisons more difficult than they need to
 ↪ be.

What makes the comparison difficult?
The density curves are somewhat redundant.

Once you've separated the groups into panels, coloring by ELS doesn't add much
 ↪ information.
```

```
The density curves may draw attention away from the histogram shapes.
The bins are fairly narrow.

The distributions look noisy, making it harder to see differences in center, spread, and
 ↪ skewness.
There are no reference markers for center.

It's difficult to quickly judge whether one group tends to have higher inflammation
 ↪ levels.
Outliers stretch the x-axis.

A few large values (especially around 10-17) compress the bulk of the data.
Option 1: Simpler faceted histograms (Recommended)
Remove the density curves and use the same bin boundaries in both panels.



ggplot(assign1, aes(x = Inflammation)) +
  geom_histogram(
    aes(y = after_stat(density)),
    bins = 15,
    fill = "skyblue",
    colour = "black"
  ) +
  facet_wrap(~ELS) +
  labs(x = "Inflammation",
       y = "Density") +
  theme_bw()
This allows viewers to focus on the histogram shape itself.

Option 2: Add group means or medians
This is often the most useful improvement when the research question concerns group
 ↪ differences.



library(dplyr)

means <- assign1 %>%
  group_by(ELS) %>%
  summarise(mean_inf = mean(Inflammation, na.rm = TRUE))

ggplot(assign1, aes(x = Inflammation)) +
  geom_histogram(
    aes(y = after_stat(density)),
    bins = 15,
    fill = "lightblue",
    colour = "black"
  ) +
  geom_vline(data = means,
             aes(xintercept = mean_inf),
             colour = "red",
             linewidth = 1.2) +
  facet_wrap(~ELS) +
  theme_bw()
```

```
The vertical line immediately communicates whether one group has higher average
 ↪  inflammation.

Option 3: Use transparency and overlay the histograms
If your goal is specifically to compare the distributions directly, an overlay can work
 ↪  better than faceting.



ggplot(assign1,
       aes(x = Inflammation,
           fill = factor(ELS))) +
  geom_histogram(
    aes(y = after_stat(density)),
    alpha = .4,
    position = "identity",
    bins = 15
  ) +
  labs(fill = "ELS") +
  theme_bw()
Advantages:

Direct visual comparison.
Easy to see shifts in location.
Disadvantage:

Can become messy if the distributions overlap heavily.
Option 4: Faceted histograms with comparable mean/median markers
My preferred version for a report:



ggplot(assign1, aes(x = Inflammation)) +
  geom_histogram(
    aes(y = after_stat(density)),
    bins = 15,
    fill = "grey80",
    colour = "black"
  ) +
  stat_summary(
    data = assign1,
    fun = median,
    geom = "vline",
    aes(xintercept = after_stat(x)),
    colour = "red"
  ) +
  facet_wrap(~ELS) +
  theme_bw()
A median line is often preferable because your histograms appear right-skewed and include
 ↪  some large values. Medians are less influenced by extreme observations than means.

Option 5: Keep the histogram but add a rug plot


ggplot(assign1, aes(x = Inflammation)) +
```

```
  geom_histogram(
    bins = 15,
    fill = "lightblue",
    colour = "black"
  ) +
  geom_rug(alpha = .3) +
  facet_wrap(~ELS) +
  theme_bw()
The rug plot helps reveal clustering and outliers that can be hidden by binning.

For answering your substantive question
Because your research question is:

"Do participants with adverse early-life stressors have different levels of inflammation
 ↪ than those without?"

I would recommend:

Faceted histograms with identical scales (as you already have),
Add a vertical line for the group mean or median, and
Supplement with boxplots or violin plots.
In fact, a boxplot is often much better than a histogram for comparing groups:



ggplot(assign1,
       aes(x = factor(ELS),
           y = Inflammation)) +
  geom_boxplot(fill = "skyblue") +
  labs(x = "Early Life Stress",
       y = "Inflammation") +
  theme_bw()
A histogram is excellent for examining the shape of each distribution, but a boxplot
 ↪ typically makes differences in center, spread, and outliers much easier to evaluate
 ↪ when comparing two groups. In a report, I would likely present both: the histogram
 ↪ for distribution shape and the boxplot for the group comparison.




You said:
I think option 2 is great. Can I add the vline with median value, and also add label of
 ↪ the values on the plots?
Copilot said:
Absolutely. Adding a median line and labeling its value is a great enhancement,
 ↪ especially because your inflammation distributions appear somewhat right-skewed. The
 ↪ median is a robust measure of the "typical" inflammation level.

Here's one approach using dplyr and ggplot2:



library(dplyr)
library(ggplot2)
```

```
# Calculate group medians
medians <- assign1 %>%
  group_by(ELS) %>%
  summarise(median_inf = median(Inflammation, na.rm = TRUE))

ggplot(assign1, aes(x = Inflammation)) +
  geom_histogram(
    aes(y = after_stat(density)),
    bins = 15,
    fill = "lightblue",
    colour = "black"
  ) +
  geom_vline(
    data = medians,
    aes(xintercept = median_inf),
    colour = "red",
    linewidth = 1.2,
    linetype = "dashed"
  ) +
  geom_text(
    data = medians,
    aes(
      x = median_inf,
      y = 0.19,
      label = paste0("Median = ", round(median_inf, 2))
    ),
    colour = "red",
    angle = 90,
    vjust = -0.4,
    size = 4
  ) +
  facet_wrap(~ELS) +
  labs(
    x = "Inflammation",
    y = "Density"
  ) +
  theme_bw()
A potentially cleaner version
Instead of rotating the label vertically, place it near the top of the panel:



geom_text(
  data = medians,
  aes(
    x = median_inf,
    y = 0.18,
    label = round(median_inf, 2)
  ),
  colour = "red",
  fontface = "bold",
  vjust = -0.5
)
``
```

Even better: label both median and sample size
For group comparisons, reviewers and instructors often appreciate seeing both:

```
stats <- assign1 %>%
  group_by(ELS) %>%
  summarise(
    median_inf = median(Inflammation, na.rm = TRUE),
    n = n()
  )
```

Then:

```
geom_text(
  data = stats,
  aes(
    x = median_inf,
    y = 0.18,
    label = paste0("Median = ", round(median_inf, 2),
                   "\nn = ", n)
  ),
  colour = "red",
  fontface = "bold"
)
```

One small improvement to your original plot
Since the panels already separate the groups, you don't really need colour = ELS anymore.
↪ A cleaner graph is often:

```
ggplot(assign1, aes(x = Inflammation)) +
  geom_histogram(
    aes(y = after_stat(density)),
    bins = 15,
    fill = "grey80",
    colour = "black"
  ) +
  ...
```

This reduces visual clutter and makes the median lines stand out more clearly.

For a report or assignment, I'd probably use grey histograms + red dashed median lines +
↪ median labels, because the reader can immediately compare both the shape of the
↪ distributions and their central tendencies.

You said:
Sorry, I mean, I'd like to represent both mean and median.
Copilot said:

```
Yes, and I think that's actually the best choice here. Showing both the mean and median
 ↪ helps readers evaluate not only group differences, but also the degree of skewness
 ↪ within each group.

You can calculate both statistics and plot them as different colored reference lines:



library(dplyr)
library(ggplot2)

stats <- assign1 %>%
  group_by(ELS) %>%
  summarise(
    mean_inf = mean(Inflammation, na.rm = TRUE),
    median_inf = median(Inflammation, na.rm = TRUE)
  )

ggplot(assign1, aes(x = Inflammation)) +
  geom_histogram(
    aes(y = after_stat(density)),
    bins = 15,
    fill = "grey80",
    colour = "black"
  ) +

  # Mean
  geom_vline(
    data = stats,
    aes(xintercept = mean_inf),
    colour = "blue",
    linewidth = 1.2
  ) +

  # Median
  geom_vline(
    data = stats,
    aes(xintercept = median_inf),
    colour = "red",
    linewidth = 1.2,
    linetype = "dashed"
  ) +

  # Mean label
  geom_text(
    data = stats,
    aes(
      x = mean_inf,
      y = Inf,
      label = paste0("Mean = ", round(mean_inf, 2))
    ),
    colour = "blue",
    vjust = 1.5
  ) +
```

```
  # Median label
  geom_text(
    data = stats,
    aes(
      x = median_inf,
      y = Inf,
      label = paste0("Median = ", round(median_inf, 2))
    ),
    colour = "red",
    vjust = 3
  ) +

  facet_wrap(~ELS) +
  theme_bw()
Interpretation
This figure becomes quite informative:

Blue solid line = group mean
Red dashed line = group median
When:

Mean  Median → distribution is fairly symmetric.
Mean > Median → positive (right) skew.
Mean < Median → negative (left) skew.
Looking at the histogram you attached, I would expect the mean to be somewhat larger than
 ↪ the median in both groups because of the long right tails and a few large
 ↪ inflammation values.

An even cleaner alternative
Instead of placing labels at the top where they can overlap, combine them into a single
 ↪ annotation:



stats <- assign1 %>%
  group_by(ELS) %>%
  summarise(
    mean_inf = mean(Inflammation, na.rm = TRUE),
    median_inf = median(Inflammation, na.rm = TRUE)
  ) %>%
  mutate(
    label = paste0(
      "Mean = ", round(mean_inf, 2),
      "\nMedian = ", round(median_inf, 2)
    )
  )
then add:



geom_text(
  data = stats,
  aes(x = Inf, y = Inf, label = label),
  hjust = 1.1,
  vjust = 1.1,
```

```
  inherit.aes = FALSE
)
This places a small summary box in the upper-right corner of each facet and is often
 ↪  easier to read in a report or assignment.
```

```
Add sample sizes to each facet

Adjust label placement automatically

Add confidence intervals for means
```